

GeoAgentic-RAG: A Multi-Agent framework for autonomous geospatial reasoning and visual insight generation with LLM

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ABSTRACT

Conventional Retrieval-Augmented Generation (RAG) systems have limited effectiveness in geospatial question answering because text-based similarity retrieval cannot adequately represent spatial semantics such as topology and spatial context. To overcome this limitation, we propose GeoAgentic-RAG, a multi-agent framework that enables multimodal large language models (MLLMs) to perform autonomous geospatial reasoning. The framework integrates natural language query parsing, semantic-spatial retrieval, and executable geospatial analysis within a unified, agent-based workflow. Multiple specialized agents collaboratively interpret user queries, retrieve relevant vector and raster datasets from a unified geospatial database, decompose analytical tasks, generate valid spatial logic, and produce interpretable analytical results. We evaluate GeoAgentic-RAG using a benchmark of geospatial retrieval, feature characterization, and spatial relational reasoning tasks in Nanjing and Guangzhou. The proposed framework achieves a pass rate of 85.3% and an answer correctness of 88.3%, outperforming conventional RAG methods and representative code-generation baselines. These results demonstrate that agent-based integration of retrieval and spatial analysis substantially improves the reliability of geospatial question answering and provides a practical framework for the next-generation intelligent GIS applications.

1. Introduction

The rapid advancement of artificial intelligence (AI), particularly the progress of Large Language Models (LLMs) in fields such as Natural Language Processing (NLP) and information retrieval (Arslan et al., 2024) has introduced new opportunities for data retrieval and generative reasoning. Among recent innovations, Retrieval-Augmented Generation (RAG) has become a central approach for improving semantic comprehension and factual consistency of generative models (Edge et al., 2025; Vidivelli et al., 2024). RAG combines external knowledge retrieval with text generation, allowing models to incorporate relevant information from external databases or document repositories during response generation. This integration improves both the accuracy and

coherence of generated output. As a result, it has become a central approach for strengthening the semantic reasoning capability of large-scale language models (Arslan et al., 2024).

Geographic Information Systems (GIS) provide the computational foundation for acquiring, managing, analysing, and visualizing spatial data (Syafiq and Azri, 2023). They support a wide range of applications, including smart city planning (Lee and Yang, 2025), environmental monitoring (Zhao et al., 2024), natural hazard management (Maleki and Mohammadalizadehkorde, 2022), community commercial vitality evaluating (Cui et al. 2025) and urban traffic control (Wang et al., 2023). Spatial data differ substantially from textual or visual data because they contain multi-dimensional and heterogeneous elements, including geographic coordinates, topological relations, geometric structures, and

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remote sensing rasters. This complexity far exceeds that of conventional unstructured data (Zhu and Zhou, 2021). However, existing RAG frameworks have been developed primarily for linguistic and visual domains and do not model spatial relationships, geographic semantics, or spatiotemporal contexts. As a result, they perform poorly in tasks such as spatial question answering, geographic retrieval, and spatial content generation. When spatial operations such as spatial overlay, context fusion, and topological analysis are required, their performance deteriorates further, constraining the application of RAG in geospatial intelligence (Li and Ning, 2023).

Integrating RAG into GIS research therefore represents both an extension of NLP methods and a potential shift in how spatial semantics and geospatial reasoning are approached. RAG technology offers three key advantages in this context. First, it supports natural language-driven geospatial question-answering (Geospatial QA), allowing users to query and reason over spatial data without explicit knowledge of GIS commands or syntax. Second, its retrieval mechanism can incorporate heterogeneous spatial sources, such as remote sensing imagery, vector entities, and geographic knowledge bases, into a unified space, enabling the development of semantically enhanced spatial models. Third, the generative component of RAG can translate analytical results into interpretable, context-aware linguistic responses, providing a natural interaction layer for geospatial decision support systems. These capabilities offer a technical foundation for the continued evolution of GIS toward higher levels of intelligence, interoperability, and accessibility.

Despite these prospects, major challenges remain. Existing RAG systems lack dedicated spatial indexing and spatial parsing modules capable of representing spatial proximity, containment, and connectivity. Similarly, current language models are limited in their ability to capture spatiotemporal correlations and semantic constraints among geographic entities. They also lack the ability to interpret map semantics, layer hierarchies, and spatial logic. Addressing these issues requires a hybrid architecture that integrates spatial perception, geographic reasoning, and language generation within a single operational framework in order to link LLMs more directly with GIS.

To meet this need, this study introduces GeoAgentic-RAG, a multi-agent RAG framework designed for geospatial semantic reasoning and autonomous analytical processing. In contrast to conventional systems, GeoAgentic-RAG can identify required datasets automatically, retrieve relevant GIS resources from local or distributed databases, and perform visualization and analytical tasks without manual specification of data paths or operations. Through multi-agent collaboration and context-driven task allocation, the framework integrates structured GIS data and natural language inputs to support complex spatial problem solving, semantic retrieval, and geographic context generation. It represents the first attempt to embed spatially aware agents within a RAG architecture, creating a closed-loop workflow that spans data retrieval, analysis, and visualization stages. This framework establishes a new technical paradigm for the integration of GIScience and large language models.

This paper presents the design principles, core components, and technical architecture of GeoAgentic-RAG, and evaluate its performance across several geospatial retrieval and reasoning tasks. Experimental results demonstrate that the framework improves the efficiency, accuracy, and interpretability of geographic information processing. These findings indicate that GeoAgentic-RAG provides an effective and scalable solution for intelligent geospatial applications, including smart city governance, environmental monitoring, and disaster management.

2. Related works

RAG combines information retrieval with natural language generation to enhance the factual and contextual accuracy of large language models. In its standard architecture, a retrieval module searches for documents relevant to a user query, and a generation module produces responses conditioned on both the retrieved content and the query itself (Arslan et al., 2024; Li et al., 2022). This hybrid design allows generative

models to address knowledge-intensive tasks that exceed the scope of their pre-trained parameters. Conventional LLMs rely solely on static corpora; once training is completed, their parameters remain fixed. Consequently, they perform poorly when confronted with time-sensitive information or domain-specific knowledge. For instance, an LLM may return outdated or inaccurate results when asked about newly published research or rapidly evolving technical fields. RAG mitigates these shortcomings by retrieving up-to-date information from external databases or literature repositories and integrating it into the generation process (Bao et al., 2025). In practice, the system identifies several relevant text passages, combines them with the user query to form an augmented prompt, and passes the result to the generator to produce contextually consistent answers. This procedure improves factual grounding and supports complex reasoning tasks as long-form question answering and multi-hop reasoning (Iskan et al., 2024).

Early implementations of RAG are confined largely to text generation tasks, including question answering, text summarization, and machine translation. These studies establish the foundation for its widespread adoption in natural language processing. As research progressed, various extensions have been proposed to improve adaptability and efficiency (Li et al., 2022). Adaptive-RAG (Jeong et al., 2024) employs a dynamic parser that adjusts retrieval strategies according to question complexity, maintaining balance between computational efficiency and semantic coverage. TableGPT (Zha et al., 2023) adapts RAG to structured data by translating table content into SQL queries, enabling databases to be accessed through natural language instructions. Other work focuses on text-parsing modules and document loaders that transform heterogeneous knowledge sources into uniform textual input, improving retrieval accuracy. A persistent difficulty in these models is the occurrence of “hallucinations”, where generated text deviates from the retrieved facts (Hager et al., 2024). GraphRAG (Edge et al., 2025) addresses this issue by introducing graph-based indexing and annotated representations that strengthen semantic accuracy between retrieved and generated content.

With the maturation of the field, RAG has expanded beyond pure text to multi-modal applications. In computer vision, RA-CM3 (Yasunaga et al., 2023) integrates image understanding with textual reasoning (Bai et al., 2023), marking one of the first multimodal RAG architecture. BLIP-2 (Li et al., 2023) combines frozen image encoders with LLMs to achieve zero-shot image-to-text translation. The “Visualize before you write” approach (Zhu et al., 2023) further demonstrates how generated imagery can guide text production in open-ended scenarios. Comparable advances appear in audio and video processing: the GSS framework stitches audio segments to align speech and translation data (Zhao et al., 2023), while VisRAG (Yu et al., 2025) applies vision-language models to document imagery, achieving 20–40% gains in end-to-end performance over text-only RAG. These developments indicate the potential of RAG as a general strategy for reasoning across multiple data modalities.

Parallel to these advances, GIS have become indispensable for managing and analysing spatial data. GIS datasets contain not only geometric and locational information but also networks of spatial relationships and semantic attributes. Such characteristics are central to applications in urban planning, environmental assessment, disaster early warning, and intelligent transportation. Integrating LLM with GIS can enhance the interpretation of geospatial data and introduce natural language interaction into spatial analysis workflows. Recent studies have begun exploring this integration.

The Geo-GPT framework (Zhang et al., 2024) applies LLMs to geospatial tasks by introducing agent mechanism capable of calling search engines, map APIs, and remote sensing interfaces. This design enables natural language instructions to trigger spatial data retrieval and analysis, supporting intelligent applications in urban governance and resource monitoring. Building on this idea, GeoTool-GPT (Wei et al., 2025) fine-tunes the LLaMA-2-7B model to interface directly with GIS tools such as clipping and intersection. The framework constructs a “task-tool-call” mapping dataset and integrates tool knowledge,

allowing the model to execute operations like map drawing, spatial querying, and buffer analysis through natural language prompts. Similarly, LLM-Geo (Li and Ning, 2023) positions the language model as the reasoning core for automatic spatial data analysis and visualization. By combining natural language understanding, logical reasoning, and code generation, it can formulate geoprocessing workflows that address user queries automatically. Peng et al. (Peng et al., 2023) extend this direction by employing database-based and vector-similarity retrieval methods to improve the interactivity of spatial vector data. In sum, these studies highlight the growing interest in linking LLM with GIS industry for intelligent spatial applications.

Despite these achievements, existing RAG and GIS-LLM approaches still face substantial limitations. Most RAG systems are optimized for unstructured text or imagery and lack mechanism to interpret the spatial attributes intrinsic to GIS data, such as coordinate reference systems, projection transformations, and spatial indexing. As a result, they cannot incorporate geospatial data effectively into external knowledge bases. The coexistence of multiple data types, including vector and raster, also

poses challenges for multi-modal data fusion, and current RAG models remain inadequate for integrating heterogeneous spatial information. Furthermore, GIS-LLM frameworks struggle with complex analytical workflows. For example, LLM-Geo requires the generation of long procedural scripts for spatial analysis, yet these scripts are prone to error and lack automated correction. Many systems also depend on explicit user instructions to specify required datasets, limiting autonomy in data identification and retrieval. Finally, the continuous growth of spatio-temporal big data demands dynamic updating and efficient processing, which most current architectures cannot yet provide.

In summary, although notable progress has been made in linking LLMs and GIS, existing research remains largely exploratory. There is still no comprehensive framework capable of fully exploiting the spatial reasoning and analytical potential of language models or achieving seamless integration with GIS. The development of such a framework is essential for advancing geospatial intelligence toward greater automation, adaptability, and interpretability.

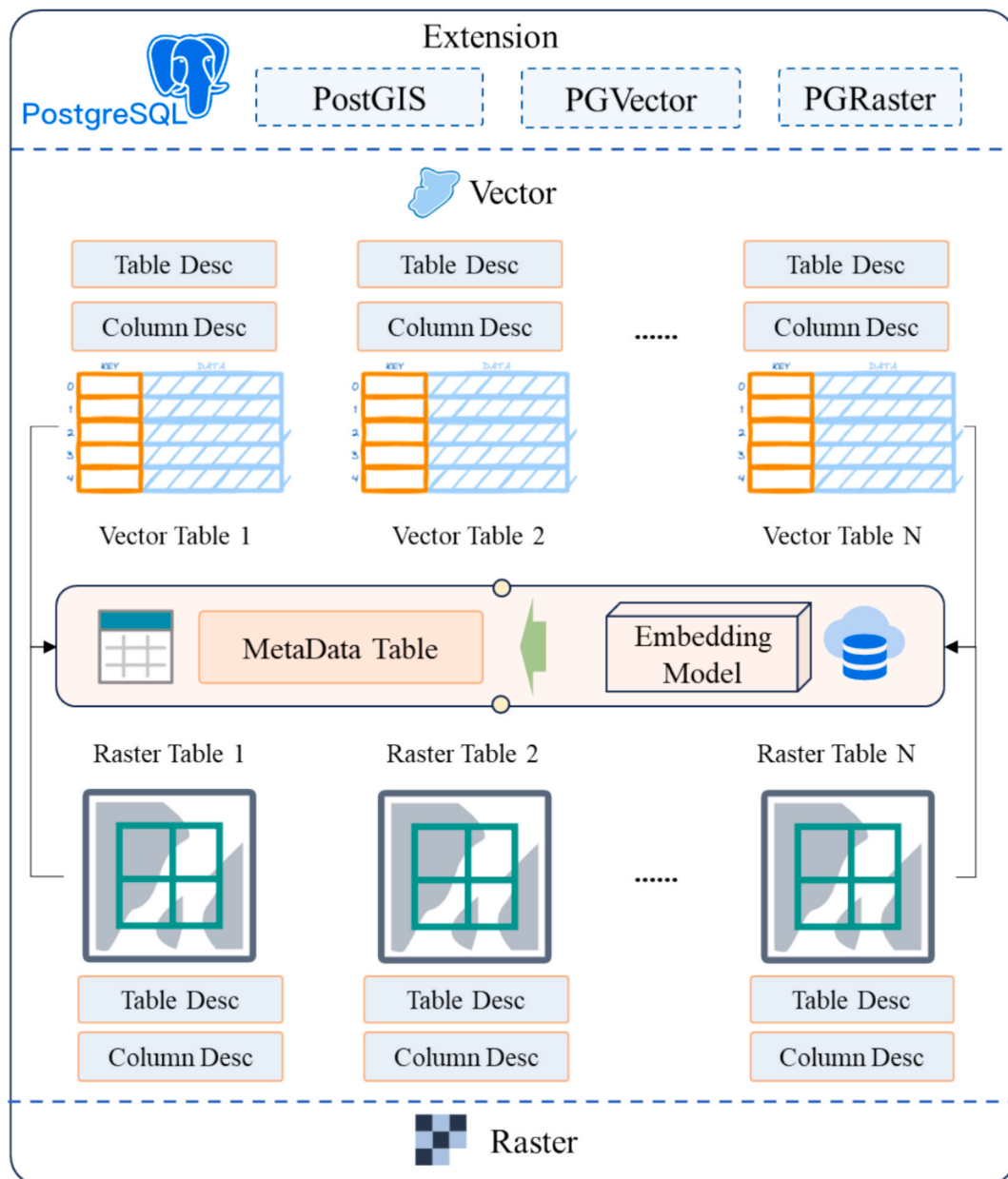


Fig. 1. Unified Geospatial Data Preprocessing Workflow.

3. Methodology

We propose GeoAgentic-RAG, a multi-agent framework for automated geospatial question answering that integrates natural language interaction, semantic-spatial retrieval, and executable geospatial analysis. The system processes user queries by decomposing them into atomic spatial tasks, retrieving relevant vector or raster datasets from a standardized PostGIS-based knowledge base, and assigning subtasks to specialized agents. Each agent operates under a ReAct-style interaction implemented with LangGraph, using GPT-4 as the core reasoning model. The framework supports fully automated, end-to-end workflows, from query interpretation and data retrieval to spatial analysis and result visualization, without manual specification of datasets or analytical procedures.

3.1. Data Standardization and Preprocessing

Efficient storage, management, and retrieval of geospatial data are essential for autonomous operation. GeoAgentic-RAG employs a unified data management framework based on PostGIS (Fig. 1), which supports both vector geometries (points, lines, polygons) and raster datasets (e.g., satellite imagery, Digital Elevation Model) through the PGRaster extension. Each dataset is stored in its native spatial format and augmented with a concise natural language description summarizing its thematic content, spatial extent, temporal coverage, and functional role (e.g., land-use classification or road network structure). These descriptions function not only as metadata but also as the semantic interface linking raw geospatial data with language models.

To support semantic-aware retrieval, all dataset descriptions are encoded as dense vector embeddings using the Qwen3-Embedding-8B model (Zhang et al., 2025). The resulting embeddings are stored in a PGVector-enabled table, allowing efficient cosine similarity search in high-dimensional space. This design allows downstream agents to retrieve semantically relevant and structurally valid geospatial datasets in response to natural language queries, thus bridging unstructured user input and structured geospatial databases.

3.2. GeoAgentic-RAG multi-agent framework

The core of GeoAgentic-RAG framework lies in its multi-agent-driven RAG mechanism, which achieves full automation from natural language input to task decomposition, data retrieval, and result generation. Multiple specialized agents collaborate to process information across successive stages of geospatial analysis, enabling a fully autonomous and coherent workflow. The multi-agent architecture is implemented using *LangGraph*, with GPT-4o serving as the core reasoning engine. Each agent is designed using the ReAct prompt-engineering approach (Yao et al., 2023), allowing it to integrate retrieval knowledge with generation strategies dynamically during execution.

Through coordinated interaction among agents, GeoAgentic-RAG perform dynamic task scheduling and allocation for complex geospatial information processing. This collaborative mechanism enhances the efficiency of spatial data retrieval and integration while improving the accuracy and interpretability of analytical results (Fig. 2). The multi-agent design also offers scalability and robustness, enabling stable operation under heterogeneous, multi-source, data conditions and intricate spatial logic.

3.2.1. Natural language-driven task parsing and data exploration

The natural language-driven task parsing and data exploration module is the starting point for autonomous geospatial analysis. Its purpose is to translate user queries or analytical requests, expressed in natural language, into structured task representations that guide subsequent retrieval and processing. The module interprets user intent, spatial scope, temporal coverage, and data requirements, enabling automatic mapping of problems to data sources explicit user direction.

Semantic analysis decomposes the query into independent subtasks. During this process, the model identifies the geographic entities involved, such as roads, rivers, buildings, or land-use types, and determines the required data types, including vector, raster, or auxiliary multimodal information. Based on the extracted semantic features of the user query, the system then constructs data exploration strategies to identify relevant geospatial datasets. Candidate data sources are ranked according to spatial indexes, attribute relevance, and spatiotemporal constraints.

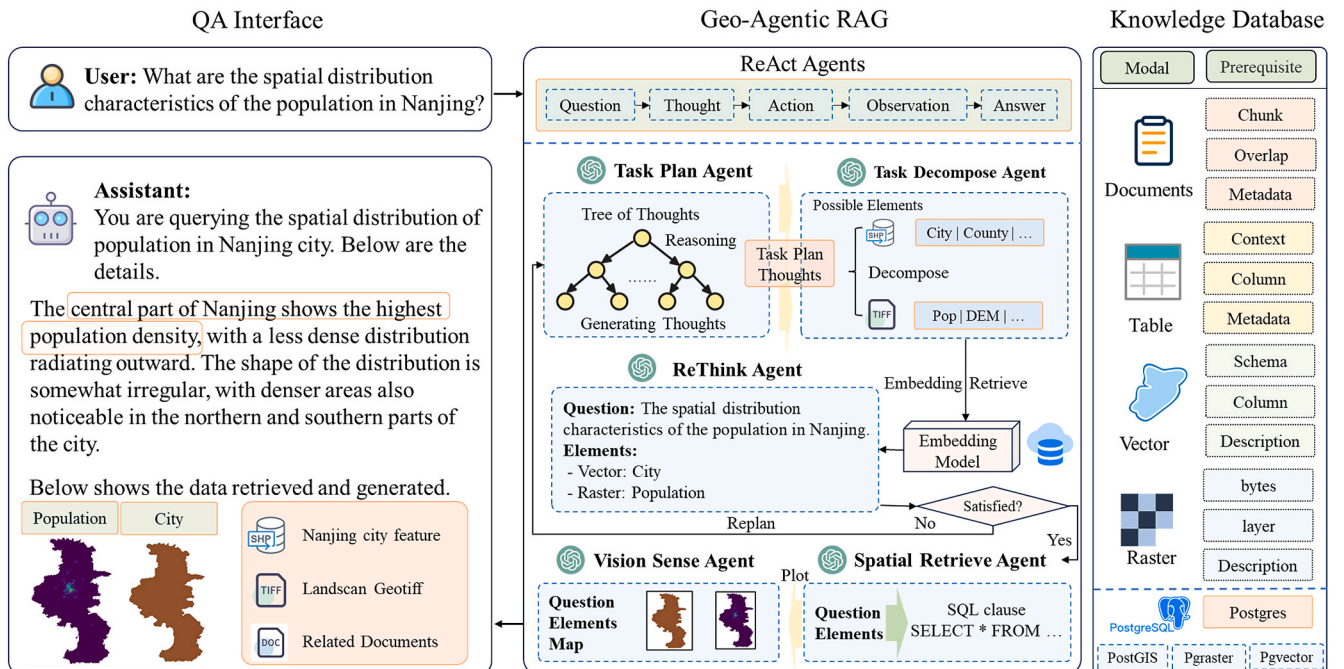


Fig. 2. Framework diagram of the GeoAgentic-RAG framework.

Formally, a user query Q is first decomposed into a set of atomic geospatial subtasks:

$$Q = \{q_1, q_2, \dots, q_n\} \quad (1)$$

where each subtask q_i represents a specific spatial operation or condition. A mapping function $f_{\text{mapper}}(Q)$ is then applied to associate the set of subtasks with structured geospatial data entries:

$$T = \{t_1, t_2, \dots, t_m\} = f_{\text{mapper}}(Q) \quad (2)$$

Where each t_i represents a candidate geographic dataset to be retrieved.

For semantic similarity matching between query representations and dataset descriptions, Cosine Distance is adopted as the similarity metric (Steck et al., 2024):

$$d_{\text{cos}}(A, B) = 1 - \frac{A \cdot B}{\|A\| \|B\|} \quad (3)$$

Here A denotes semantic vector of the target geographic entity and B the vector representation in the database. Smaller cosine distance indicates greater semantic similarity, with 0 denoting perfect similarity and 2 denoting complete dissimilarity.

This module improves both data retrieval efficiency and relevance by constructing a mapping between natural language queries and spatial datasets. It enables GeoAgentic-RAG to complete the transition from problem understanding to data discovery autonomously, forming the foundation for downstream reasoning and visualization.

3.2.1.1. Spatial task thinking and analysis in multi-agent collaboration. Within GeoAgentic-RAG, the multi-agent collaboration mechanism

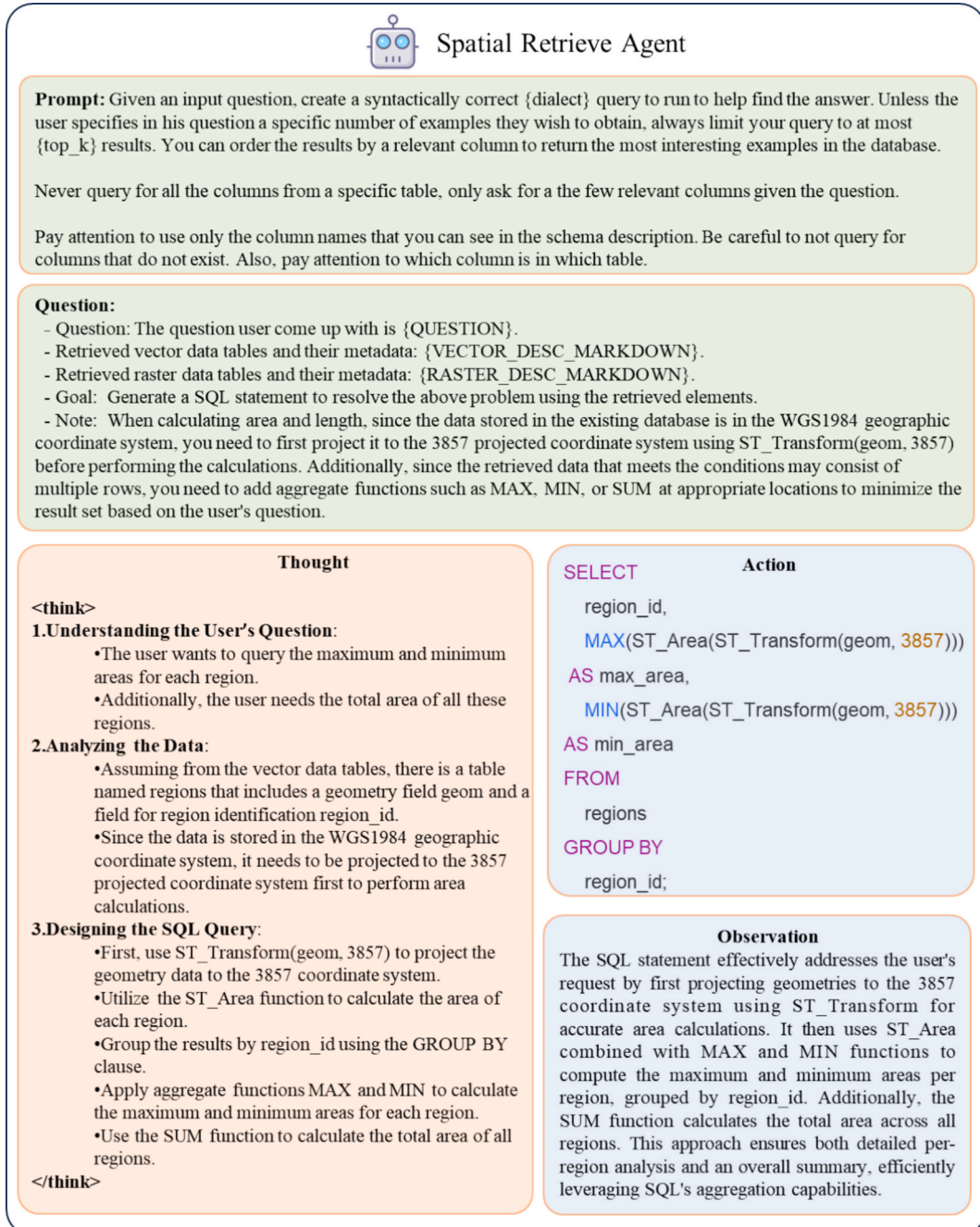


Fig. 3. Example of the "Thought-Action-Observation" Loop in a Spatial Query Task.

performs distributed execution and reasoning for spatial tasks. Unlike single model systems, it employs a heterogeneous network of agents capable of semantic-level information exchange and collective reasoning. Collaboration is organized automatically based on task semantics, eliminating the need for manual supervision. The mechanism supports the complete workflow, from data acquisition and spatial computation to analytical output, under an integrated reasoning framework.

3.2.1.1.1. Construction of task dependency graph and hierarchical execution. The structured sequence of subtasks produced by the parsing module is analysed for dependencies and represented as a *directed acyclic graph* (DAG):

$$G = (V, E) \quad (4)$$

where V represents the set of task nodes and E denotes dependency edges. A directed edge (t_i, t_j) indicates task t_j requires the output of task t_i . The DAG determines execution order and parallelization, allowing tasks to be partitioned and executed hierarchically across multiple agents.

GeoAgentic-RAG supports dynamic updates of the task graph. If the output of a subtask is incomplete or inconsistent, the system automatically adjusts the graph and replans execution. This feature provides fault tolerance and flexibility for complex spatial analysis.

3.2.1.1.2. Role differentiation and functional collaboration of multi-agents. To support efficient execution of heterogeneous spatial operations, GeoAgentic-RAG adopts a role-differentiated multi-agent architecture based on the ReAct (Fig. 3): (1) **Data Retrieval Agent**, responsible for querying and retrieving vector and raster datasets from the PostGIS database and, when required, from external geospatial knowledge sources; (2) **Spatial Analysis Agent**, responsible for executing geometric and raster-based operations, including buffering, overlay, aggregation, and clipping; (3) **Reasoning Agent**, responsible for integrating intermediate analytical outputs, performing semantic synthesis and logical inference, and generating final responses. (4) **Supervisor Agent**, responsible for monitoring workflow execution, coordinating agent interactions, resolving conflicts, and ensuring overall task consistency. These agents form a hierarchical collaboration structure in which lower-level agents handle raw data manipulation and higher-level agents conduct semantic reasoning and decision synthesis. The design enables a progressive transition from data-driven to knowledge-driven processing.

3.2.1.1.3. Message passing and joint reasoning among agents. During collaborative execution, GeoAgentic-RAG agents exchange information through shared communication interfaces. The transmitted content includes data descriptions, execution outputs, spatial relationships, and task status information. Let $A = \{a_1, a_2, \dots, a_k\}$ represent the set of agents in the system. The process of message passing between any two agents is defined as:

$$m_{i \rightarrow j} = f_{comm}(o_i, \pi_j) \quad (5)$$

where o_i denotes the output generated from agent a_i ; π_j is the task intent of the receiving agent a_j ; and f_{comm} represents the message transformation function, which converts the result into a semantic format interpretable by the receiving agent.

Through continuous message exchanges, a collaborative reasoning chain emerges among agents. This chain enables the system to refine intermediate outputs iteratively and to minimize both semantic bias and spatial error at the global level. The mechanism formalizes inter-agent communication and ensures logical consistency and geospatial convergence across all agent outputs. Consequently, they system can effectively handle ambiguous, incomplete, or multi-layer spatial problems that require cross-domain reasoning.

In summary, the multi-agent collaboration mechanism in GeoAgentic-RAG automates both the division of labour and the logical

coordination necessary for complex geospatial tasks. By integrating semantic reasoning with spatial constraints, it forms an intelligent reasoning environment distinguished by high autonomy, interpretability, and reliability. This architecture supports subsequent geographic knowledge generation and enable context-aware task decomposition, planning and tool selection within the workflow.

4. Experiments and evaluation

4.1. Evaluation metrics

To evaluate the performance of GeoAgentic-RAG in geospatial question answering, we adopt a multi-dimensional evaluation framework that emphasizes executable correctness and result fidelity. The primary evaluation metrics are Pass Rate (PR) and Answer Correctness (AC), both assessed through direct execution against the PostGIS backend.

PR measures the proportion of model-generated executable spatial programs (ESPs) that execute successfully without syntactic or semantic errors:

$$PR = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\text{Execute}(q_i^{\text{gen}}) = \text{Success}) \quad (6)$$

where N is the total number of test queries, q_i^{gen} denotes the model-generated query for sample i , and $\mathbb{I}(\cdot)$ is the indicator function.

AC evaluates whether the execution output of a generated query exactly matches the manually verified ground-truth result, conditioned on successful execution.

$$AC = \frac{1}{N_{\text{exec}}} \sum_{i \in \varepsilon} \mathbb{I}(\text{Output}(q_i^{\text{gen}}) = \text{Output}(q_i^{\text{gt}})) \quad (7)$$

where ε is the set of successfully executed queries ($|\varepsilon| = N_{\text{exec}}$), and q_i^{gt} denotes the ground-truth query for sample i .

To provide a unified end-to-end measure, we defined a Hybrid Score (HS) as the product of execution success and answer correctness:

$$HS = PR \times AC \quad (8)$$

For tasks involving descriptive generation, such as spatial feature characterization, we additionally evaluate semantic quality using a Relevancy score. Relevancy is defined as the average cosine similarity between embedding representations of the model-generated responses and the corresponding reference descriptions. This metric is computed only for Spatial Feature Characterization tasks; all other task categories are evaluated exclusively using PR and AC. This dual-track evaluation design prioritize executable validity and answer correctness while incorporating semantic assessment where natural language generation is required.

4.2. Case study experimental design

To evaluate the generalization capability and practical applicability of the GeoAgentic-RAG framework, an experimental system is established that encompasses three representative task types: Geospatial Retrieval, Spatial Feature Characterization, and Spatial Relational Reasoning. These tasks are sequentially related, increasing in complexity from basic data retrieval to semantic reasoning. Together, they constitute a comprehensive assessment framework covering the full process of “understanding—retrieval—generation—reasoning”.

Experiments are conducted using geospatial datasets from two Chinese cities, Nanjing and Guangzhou. All spatial data are stored in a PostGIS-enabled PostgreSQL database, which supports integrated vector and raster storage and complex spatial operations.

4.2.1. Dataset description

The experimental datasets includes both vector and raster data types, representing diverse geographic entities and attributes. Vector datasets

consist of urban road networks, land use data, administrative boundaries, building footprints, and water bodies. Raster datasets include DEM, population density, slope, aspect, and surface roughness. These datasets represent spatial form, terrain characteristics, and human activity patterns and provide the data basis for evaluating the system's multi-modal geospatial capabilities (Table 1). They support multimodal retrieval, spatial computation, and semantic interpretation across physical, infrastructural, and socio-demographic dimensions.

4.2.2. Task design

Each task type is instantiated using natural-language queries that reflect common geospatial inquiry scenarios. For each query, a ground-truth executable spatial program (ESP) and a reference answer are manually annotated to enable quantitative evaluation.

4.2.2.1. Geospatial retrieval task. This task evaluates the framework's ability to map natural-language descriptions of geographic entities or spatial features to corresponding entries in the geospatial database. Given a query containing geographic names, feature types, or spatial constraints, the system is required to perform semantic parsing, generate the corresponding ESP, and retrieve the relevant datasets automatically. Performance reflects the efficiency of language-to-spatial-semantic conversion and the precision of multimodal retrieval. It verifies the core functionality of GeoAgentic-RAG in linking textual semantics with spatial data localization.

4.2.2.2. Spatial feature characterization task. This task measures the model's ability to extract and summarize spatial structures or distribution patterns and to express them in natural language form. The framework generates semantic descriptions based on retrieved vector or raster data, for example: "Population density in this area follows a core-periphery decreasing pattern." or "Elevation rises gradually from the southwest toward the northeast." The task assesses competence in pattern recognition, spatial summarization, and semantic coherence, representing a transition from "data-driven" to "cognition-driven" geospatial intelligence.

4.2.2.3. Spatial relational reasoning task. The third task examines how the system models and infers topological relations such as proximity, inclusion, intersection, aggregation, and buffering. Example queries include "Which roads cross Qinhua district?" or "Is there a water body

Table 1
Metadata and Inventory of Experimental Datasets.

Data Type	Elements	Fields	Description For Short
Vector	County	Geom, County	Administrative boundaries for demographic and planning analysis
	Buildings	Geom, Floors	Building footprints for 3D modelling and urban structure analysis.
	Built-up Area	Geom, Year	Urbanized area boundaries for growth and land-use intensity analysis.
	Green Space	Geom, Name	Vegetated areas for ecological and green space analysis.
	Road	Geom, Category	Road network for transportation and mobility analysis.
	Water	Geom, Name	Rivers and water bodies for hydrological and environmental analysis.
	Population	Rast, layer	Population density raster for temporal trend analysis.
Raster	Aspect	Rast, layer	30 m raster representing slope aspect for solar and terrain analysis
	Slope	Rast, layer	30 m raster representing slope steepness for terrain analysis.
	DEM	Rast, layer	30 m DEM
	Surface Roughness	Rast, layer	30 m raster representing terrain roughness for aerodynamic and hydrological analysis.

within 500 m of the industrial zone?" This task evaluates the framework's ability to perform topological reasoning, maintain semantic consistency, and integrate cross-modal spatial knowledge.

4.3. Baselines and ablation

To evaluate the contribution of GeoAgentic-RAG, we compare it against a set of representative baseline methods, including pure LLM, standard retrieval-augmented generation, and tool-augmented approaches.

4.3.1. Baselines

We consider the following baseline methods: (1) Pure LLM: GPT-4o is prompted using the original question only, with instructions to "answer based on your knowledge". No external data retrieval or tool access is provided. This setting reflects performance based solely on pretrained knowledge; (2) Standard RAG: A conventional RAG pipeline retrieves the top-k = 5 text segments from layer metadata and attribute tables using cosine similarity. The retrieved text is appended to the prompt, and GPT-4o generates a free-form response without structured query synthesis or tool invocation. (3) Tool-Augmented LLM: The LLM is granted access to the same PostGIS-based toolset as GeoAgentic-RAG but operates without agentic-based capabilities. The model attempts to generate a single ESP in one pass, with no iterative planning or error correction in case of execution failure.

4.3.2. Ablation settings

To quantify the contribution of key components within GeoAgentic-RAG, we further conduct ablation studies by selectively disabling individual modules: (1) Self-Reflection (SR): The error-analysis and query-revision loop is removed; (2) Multi-hop Planning (MP): Query generation is restricted to a single step, even for complex relational tasks; (3) Multimodal Router (MR): The learned tool-routing mechanism is replaced with a fixed rule-based strategy. All ablation variants use the same LLM backbone (GPT-4o) and database interface as the full model to ensure controlled and comparable evaluation.

4.4. Results and analysis

4.4.1. Overall performance

Table 2 summarizes the overall performance of GeoAgentic-RAG across two urban settings. The framework achieves an overall HS of 75.3% in Nanjing and 74.7% in Guangzhou, indicating stable performance across cities with different urban forms, data resolution, and schema configurations. Among successfully executed queries, AC consistently exceeds 90% in most task categories, indicating that spatial computations results are reliable once a valid ESP is generated.

PR represents the primary source of performance variation, ranging from 78.0% to 92.0% across task types and cities. Lower PR values are mainly associated with queries involving complex natural-language predicates, particularly those containing nested spatial constraints or implicit relational conditions, which increase the difficulty of accurate language-to-ESP translation.

Performance decreases with increasing task complexity. Geospatial

Table 2
Generalization Performance of GeoAgentic-RAG in Multiple Urban Scenarios.

City	Task Type	PR (%)	AC (%)	HS (%)
Nanjing	Geospatial Retrieval	92.0	93.5	86.0
	Spatial Feature Characterization	82.0	90.2	74.0
	Spatial Relational Reasoning	82.0	80.5	66.0
	Overall	85.3	88.3	75.3
	Geospatial Retrieval	90.0	93.3	84.0
Guangzhou	Spatial Feature Characterization	78.0	92.3	72.0
	Spatial Relational Reasoning	86.0	79.1	68.0
	Overall	84.7	88.2	74.7

retrieval tasks achieve the highest scores, followed by spatial feature characterization, while spatial relational reasoning tasks exhibit the lowest HS. This trend reflects the increasing demands placed on multi-step decomposition, precise topological expression, and implicit semantic interpretation in more complex spatial reasoning scenarios.

4.4.2. Baseline Comparison

Fig. 4 compares GeoAgentic-RAG with three baselines across PR, AC and HS. The Pure LLM baseline achieves an HS below 5%, reflecting its inability to generate executable spatial programs and its lack of grounding in external geospatial data. Standard RAG improves performance by incorporating retrieved metadata but does not support structured spatial query synthesis or tool execution, resulting in an HS of approximately 18%. The Tool-Augmented LLM further improves performance by enabling direct invocation of spatial functions, achieving an HS of 52%. However, this baseline lacks iterative planning and error correction mechanisms, leading to frequent execution failures and reduced result reliability with suboptimal PR and AC.

GeoAgentic-RAG outperforms all baselines across all metrics, achieving a PR of 85.3%, an AC of 88.3%, and an HS of 75.3%. These results indicate that access to spatial tools alone is insufficient for robust geospatial reasoning. Coordinated integration of retrieval, planning, and execution, and error recovery through agent-based orchestration is essential for handling real-world geospatial queries.

4.4.3. Ablation study

Table 3 presents the results of ablation experiment evaluating the contribution of key components within GeoAgentic-RAG's. Removing any individual module results in a measurable decline in performance, indicating that all components contribute to system robustness.

Disabling the SR module produces to the largest performance degradation, with HS decreasing by 14.1% (from 75.3% to 61.2%). This decline is most pronounced in queries involving spatial joins or buffer operations, where initial query formulations frequently omit required parameters or conditions. Without iterative error analysis and revision, such errors persist and lead to execution failures or incorrect results.

Removing MP reduces HS by 8.9% (to 66.4%), highlighting its importance for complex relational queries. In the absence of MP, multi-clause queries are forced into single-step formulations, increasing the likelihood of syntactic violations and execution errors. Disabling the MR

Table 3

Ablation study of GeoAgentic-RAG components.

Model Variant	PR(%)	AC (%)	HS (%)	ΔHS
Full GeoAgentic-RAG	85.3	88.3	75.3	—
Without SR	72.0	85.0	61.2	−14.1
Without MP	77.3	86.0	66.4	−8.9
Without MR	78.0	87.2	68.0	−7.3

results in a 7.3% reduction in HS (to 68.0%), indicating its role in effective tool selection. Without adaptive routing, inappropriate analytical functions are more frequently applied, reducing answer correctness.

In summary, the ablation results confirm that all three components are essential for reliable geospatial reasoning: SR ensures correctness via error recovery; MP enables structured decomposition; and MR improves tool-selection fidelity.

4.5. Case analysis

To provide qualitative insights into system behavior, we present three representative case studies illustrating successful multimodal analysis, complex spatial reasoning, and a characteristic failure mode.

4.5.1. Spatial feature characterization via multimodal data integration

This case evaluates the system's ability to integrate raster and vector data for spatial feature characterization (Fig. 5). The input query is: "What are the spatial distribution features of Nanjing's population?" The systems identifies the spatial scope (Nanjing) and analytical them (population distribution) and determines that both population density rasters and administrative boundary vectors are required.

The analysis pipeline includes raster clipping, administrative masking, statistical aggregation, and spatial clustering. PostGIS functions such as ST_Intersects and ST_Clip are used to isolate population grids within the city boundary. The generated output describes a center-periphery population gradient, with high densities in core districts (e.g., Gulou and Qinhuai) and lower densities in peripheral areas (e.g., Lishui and Pukou). The results are consistent with known urban structure and are visualized as a thematic map (Fig. 5).

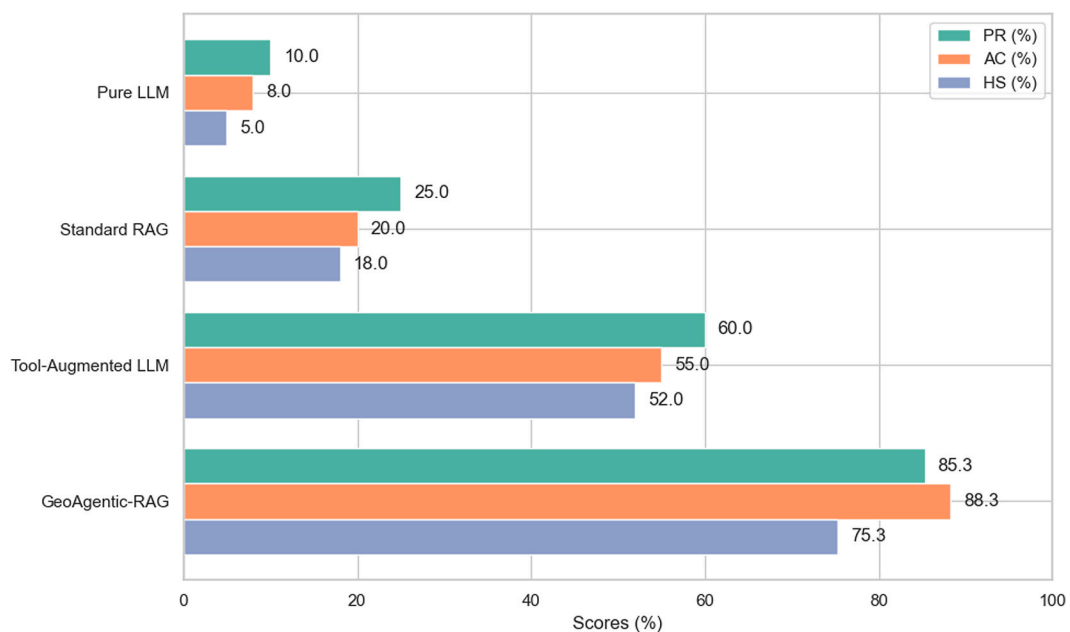
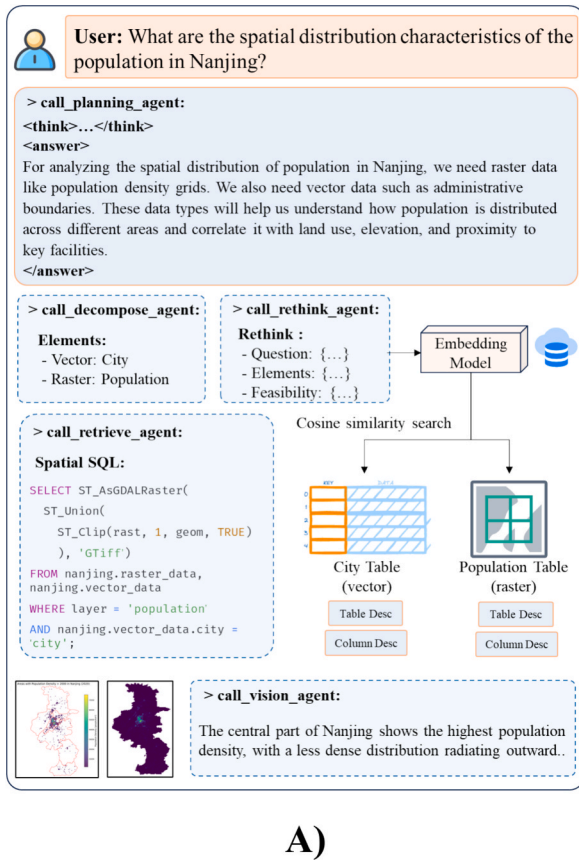
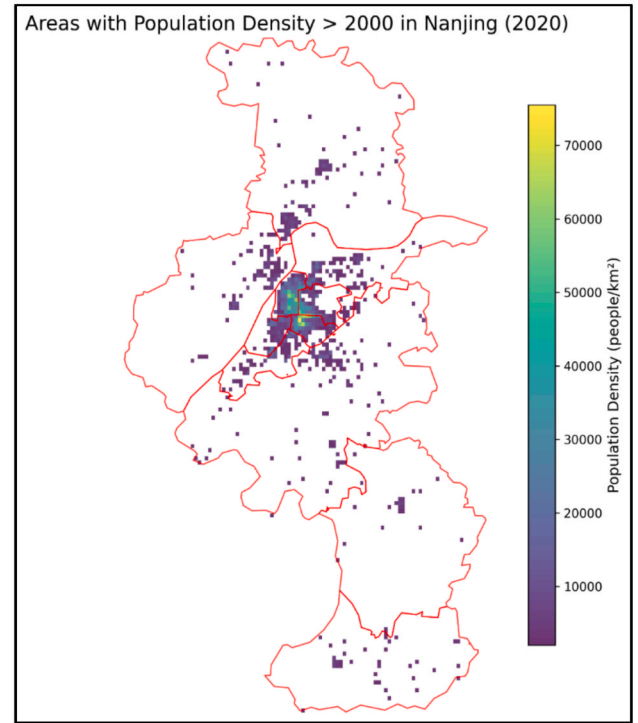


Fig. 4. Performance Comparison Across Different Metrics with Baselines.



A)



B)

Fig. 5. (A) Diagram of spatial feature analysis (B) Thematic map automatically generated by GeoAgentic-RAG.

4.5.2. Constrained spatial relational reasoning with Multi-Layer exclusion

This case demonstrates the system's ability to execute multi-layer spatial reasoning involving both vector and raster datasets. The query is defined as: "Identify potential residential development sites within Gulou District that are located within 500 m of major roads, while strictly excluding existing green spaces and areas with a slope gradient exceeding 15 degrees." It specifies residential suitability conditions based on proximity, land-cover exclusion, and terrain constraints. The system decomposes the task into a sequence of buffering, spatial subtraction, and raster filtering operations.

A buffer is applied to major roads to define accessibility zones, green spaces are excluded through polygon subtraction, and areas with slope gradients exceeding 15° are removed via raster-based filtering. The resulting ESP maintains logical consistency across heterogeneous data structures and produces a spatially coherent suitability map (Fig. 6).

4.5.3. Failure Case: Ambiguity in spatial predicate interpretation

A representative failure occurs when queries contain underspecified spatial predicates. For the query: "Identify residential buildings situated near major water bodies in high-elevation regions," the term "near" and "high-elevation" lack explicit operational definitions.

In this case, the Task Planning Agent applies default heuristic thresholds (300 m proximity and 50 m elevation). Although the generated ESP executes successfully, the resulting output is contextually inappropriate, including areas that satisfy the numeric thresholds but are not hydrologically or topographically meaningful. The Self-reflection Module does not intercept this error because the plan is internally consistent and executes without runtime exceptions.

This case highlights a limitation of current LLM-based geospatial agents: difficulty in grounding vague linguistic predicates without domain-specific knowledge or interactive clarification. While the system ensures that all results are derived from real geospatial data (avoiding

the "hallucination" common in pure LLMs), it cannot yet autonomously calibrate its spatial logic to match the implicit environmental context of a user's intent.

4.6. Efficiency and cost

We evaluate the computational efficiency and operational cost of GeoAgentic-RAG to assess its practical applicability. The framework achieves an average end-to-end latency of 12.4 s per query. Simple retrieval tasks average 6.8 s, while complex relational reasoning tasks require up to 22.6 s due to iterative planning and self-reflection.

The average token consumption per query is approximately 2,850 tokens. Although the Self-Reflection module introduces a token overhead of approximately 25%, it yields a corresponding performance gain of 14.1% in HS. By combining precomputed spatial-semantic indices with agent-based reasoning, GeoAgentic-RAG supports near-real-time execution of complex geospatial analyses that would otherwise require substantial manual GIS effort.

5. Discussion

5.1. Scientific foundation and systemic advantages of the Multi-Agent architecture

Conventional RAG frameworks, while effective for text-centric tasks, exhibit substantial limitations when applied to geospatial domains. They typically rely on static embedding spaces and textual similarity retrieval, lacking awareness of spatial semantics such as topology, directionality and scale, restricting their ability to achieve true "semantic-spatial" correlations and perform context-aware reasoning. Moreover, existing systems operate in a passive manner—retrieving information fragments without the ability to interpret geographic context, plan spatial

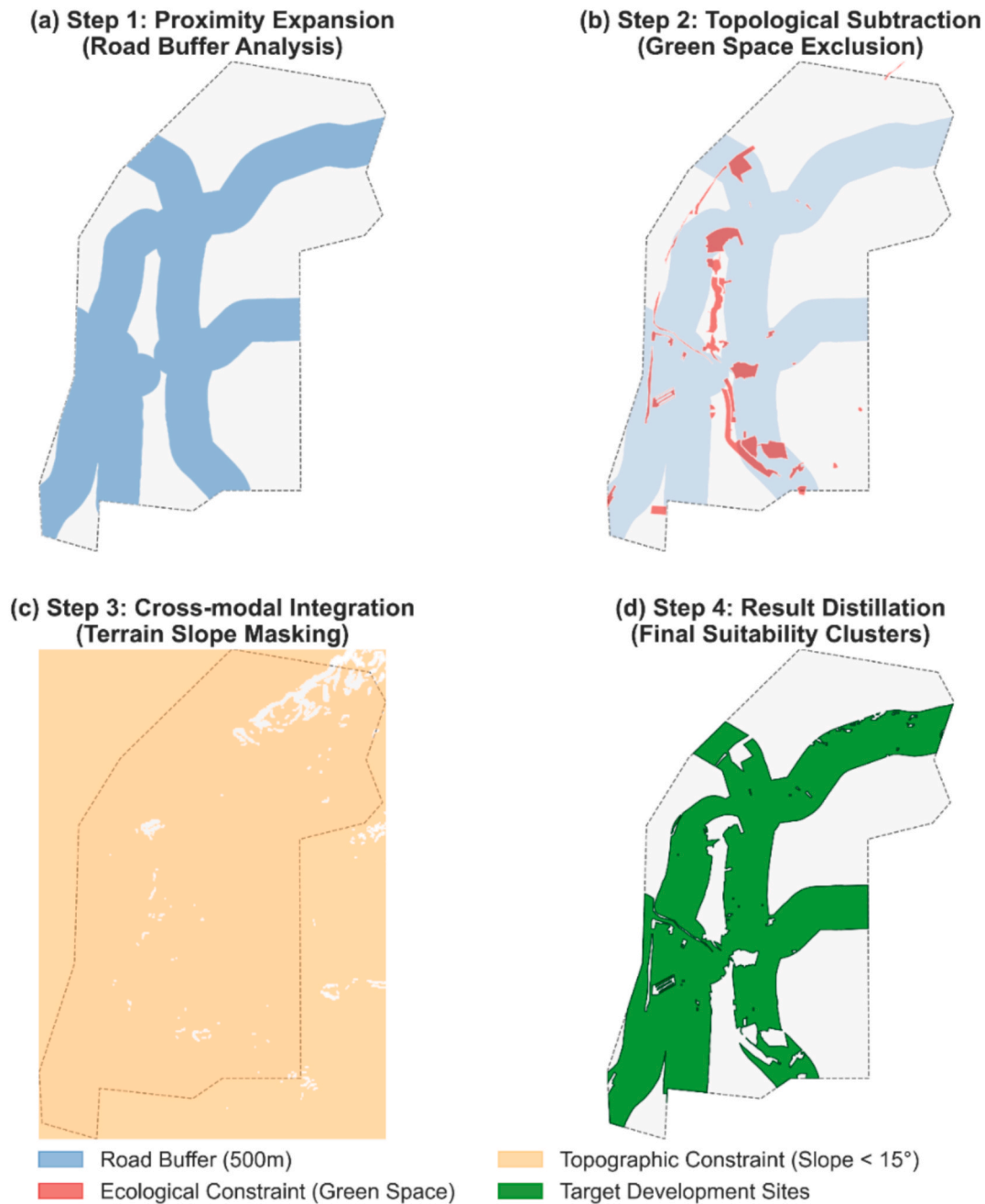


Fig. 6. Visualization of the Multi-step Geospatial Reasoning Process for Suitability Analysis.

workflows, or reason about spatial relationships. GeoAgentic-RAG overcomes these constraints through a multi-agent, spatially aware architecture that unifies natural language understanding, semantic embedding, and spatial computation. Its agent-based task parsing and vectorized spatial-semantic indexing enable autonomous identification of geographic entities, spatial relations, and temporal dependencies from use queries, generating high-fidelity ESP without manual intervention.

The principal innovation lies in bridging the semantic ambiguity of natural language with the structured characteristics of spatial databases. By combining pretrained language models with spatial indexing technologies (PostGIS + PGVector), GeoAgentic-RAG achieves context-aware, cross-modal retrieval that integrates both vector and raster data. This “semantic-driven, spatially validated” strategy enables the construction and continuous refinement of geospatial knowledge graphs. Experimental results across three task categories show that the

framework maintains high semantic consistency while ensuring factual correctness through database execution, indicating that multi-agent coordination effectively supports robust geospatial reasoning. These findings suggest that agent-based architecture provides a practical mechanism for bridging unstructured queries and structured geospatial analysis in complex spatial information processing tasks.

5.2. Geo-Semantic adaptation capability of the autonomous retrieval mechanism

Beyond automated data retrieval and visualization, GeoAgentic-RAG establishes an evolutionary pathway from data-driven to knowledge-driven geospatial intelligence. Through interaction among spatial reasoning and semantic generation agents, spatial outputs are no longer limited to geometric computations but are enriched with semantic interpretation, logical inference, and contextual understanding, forming

a closed loop of “Spatial computation – Semantic abstraction – Language generation.”.

This process contributes in two key aspects. First, the framework transforms quantitative spatial analyses into natural language narratives that retain geospatial logic, improving accessibility for users without GIS expertise. Second, by integrating vision-language mechanisms, the system extracts spatial patterns from thematic maps and expresses them in a semantic form, enabling cross-modal knowledge integration. Importantly, this semantic interpretation remain constrained by the fidelity of underlying spatial operations. When user queries contain underspecified spatial predicates (e.g., “near” or “high”), the system may generate outputs that are technically valid but contextually misaligned, reflecting an inherent challenge in open-ended geospatial question answering.

5.3. Paradigm shift, limitations and future directions

The introduction of GeoAgentic-RAG represents a paradigm shift in geographic information systems—from tool-assisted processing toward autonomous spatial cognition. Traditional GIS frameworks depend heavily on user expertise, domain knowledge, and manual operation, whereas GeoAgentic-RAG enables natural language interaction, agentic reasoning, and autonomous analytics. This “zero-code, zero-barrier” design reduces the operational burden associated with complex geospatial tasks and broadens access to advanced spatial analytics across application domains such as smart city management, natural resource monitoring, and emergency response (Fig. 7).

Despite these advantages, several limitations should be acknowledged. First, the framework relies on well-structured and semantically informative metadata, including descriptive layer summaries and attribute names. In practice, many real-world geospatial databases, particularly legacy systems, contain sparsely documented layers or cryptically named fields (e.g., pop_v2, landuse_2023), which can degrade semantic retrieval and task interpretation. Second, the current system operates without interactive clarification. When user queries contain ambiguous spatial predicates (e.g., “near a river” or “high-density area”), the framework applies predefined heuristics rather than requesting disambiguation, which may lead to contextually inappropriate results, especially in safety-critical scenarios.

The modular architecture of GeoAgentic-RAG provides a foundation for future extensions. Potential directions include incorporating spatiotemporal modeling to support dynamic analysis, integrating reinforcement learning to improve planning and tool selection, and expanding multimodal reasoning to enhance decision support in complex spatial environments. In this sense, GeoAgentic-RAG functions not only as a technical framework but also as an infrastructural step toward more intelligent and adaptive geospatial systems, while underscoring

the continued importance of human oversight in high-stakes decision-making contexts.

6. Conclusion

This study proposes and implements **GeoAgentic-RAG**, a multi-agent retrieval-augmented generation framework for geospatial tasks. The framework addresses major limitations of traditional RAG systems, namely, limited semantic understanding, weak spatial reasoning, and dependence on manual data retrieval. By integrating natural language-driven task parsing, multi-agent collaboration, and semantic-spatial fusion reasoning, GeoAgentic-RAG establishes a fully autonomous closed-loop process from query interpretation to data retrieval, analysis, and visualization, thus improving the intelligence and interpretability of geospatial processing.

Quantitative evaluation across Nanjing and Guangzhou demonstrates the effectiveness of the proposed framework. GeoAgentic-RAG achieves a pass rate of 85.3% and an answer correctness of 88.3%, substantially outperforming all baseline methods. Ablation experiments further indicate that the core components, Self-Reflection, Multi-hop Planning, and the Multimodal Router, each contribute to system robustness, with Self-Reflection alone accounting for a 14.1% improvement in the hybrid score. Qualitative case studies on population distribution characterization and constrained residential site selection illustrate the framework’s ability to execute complex, multi-layered spatial logic and to produce interpretable analytical outputs grounded in explicit spatial computation.

In summary, GeoAgentic-RAG achieves progress in four principal areas:

(1) **Semantic-Spatial Fusion Modeling:** Efficiently maps natural language to GIS data through embedding and spatial indexing.

(2) **Multi-Agent Collaborative Reasoning:** Supports automatic decomposition, parallel execution, and semantic integration of complex tasks.

(3) **Autonomous Visualization and Interpretation:** Creates a closed analytical loop from spatial computation to linguistic and visual explanation.

(4) **Low-Barrier Interaction:** Enables non-expert users to conduct sophisticated spatial analysis through natural language dialogue.

Several challenges remain. The framework is sensitive to ambiguous spatial predicates in the absence of interactive clarification and relies on semantically informative metadata for reliable retrieval and planning. Future work will focus on improving predicate grounding, incorporate spatiotemporal modelling, and further optimizing agent coordination to enhance scalability and robustness. In this respect, GeoAgentic-RAG provides a step toward infrastructure for Intelligent Geography,

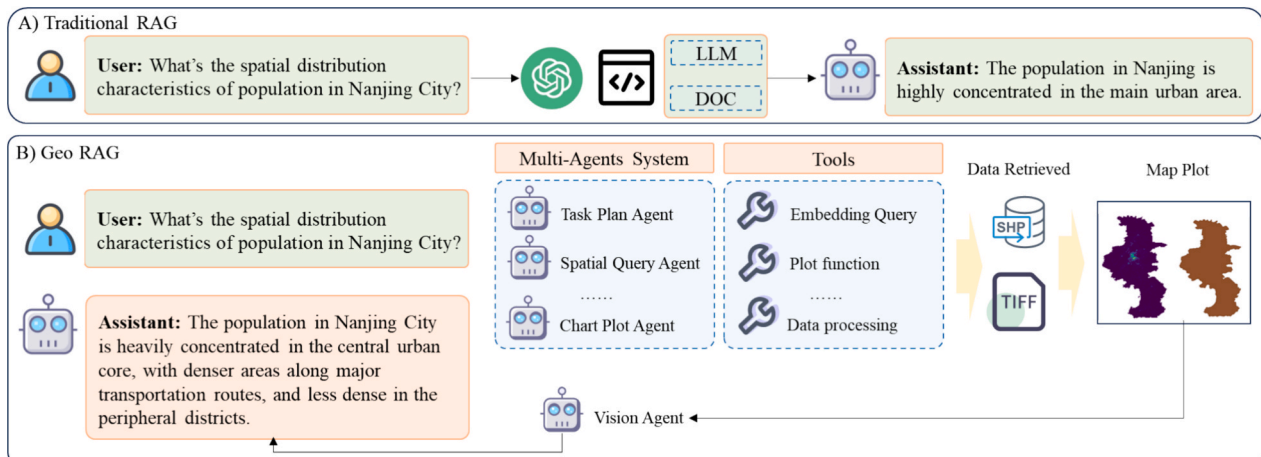


Fig. 7. Limitations of traditional RAG models in geospatial data processing.

highlighting the way for next-generation GIS systems of autonomous spatial cognition.

CRedit authorship contribution statement

Chao Liang: Writing – original draft, Formal analysis. **Yuanzheng Cui:** Writing – review & editing, Methodology, Conceptualization. **Run Shi:** Visualization, Data curation. **Guixiang Zha:** Visualization. **Xin Yin:** Investigation. **Mingzhong Xiao:** Software, Methodology. **Dong Xu:** Software. **Xuejun Duan:** Visualization, Project administration. **Bo Huang:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

<https://figshare.com/s/7c6aaab4bb0e19620f6d>

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